

Project Management Module

End-to-end construction workflow from project creation to financial reporting — mapping the MRM costing manager's framework into your ERP system

- PLANNING & SETUP
- EXECUTION & PROCUREMENT
- CERTIFICATION & BILLING
- FINANCE & PAYMENTS
- REPORTING & CONTROL
- DECISION GATE

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Project Hierarchy (WBS)

MRM's Zone → Block → Work Category → Sub-Work structure mapped to ERP

ConstructionProject model with parentProject hierarchy + isMasterProject flag



03

Create Block Sub-Projects (Gardenia, Iris, Boulevard Tower...)

Each block from MRM's spreadsheet becomes a sub-project under its zone. These are the leaf-level projects where actual work happens. `parentProject` → zone `project.isMasterProject: false`.

Project Manager
`ConstructionProject.create()`



04

Define Phases & Tasks (WBS)

For each block project, create the task hierarchy: **Phase** (level 0: Civil Works, Water Supply, etc.) → **Tasks** (level 1: Foundation, Earthwork, etc.) → **Sub-Tasks** (level 2: specific activities). This is MRM's Activity Level 3 + 4.

Costing Manager (MRM)
`ProjectTask.create()`



05

Prepare BOQ (Bill of Quantities)

Create BOQ documents (`ProjectBOQ` headers) per work category. Add line items (`BOQItem`) with quantities, unit prices, phases, categories. Items are grouped by phase (Civil, Water, etc.) and categorised by sub-work. This is MRM's core "INPUT INTERFACE — BOQ".

Costing Manager (MRM)
`ProjectBOQ + BOQItem.create()`



06

Set Budget Categories & Milestones

Define budget breakdown (Civil Works, Electrical, Plumbing, etc.) and billing milestones with trigger percentages. Budget = sum of BOQ. Milestones tie to billing events.

Costing Manager
`project.budgetCategories[]`
`project.milestones[]`



♦ Budget Approval Gate

✓ PM + Director Approve → Proceed

✗ Rejected → Revise BOQ



07

Budget Approved & Baseline Set

Budget status moves from Draft → Submitted → Approved. `totalApprovedBudget` is locked. All future actuals are compared against this baseline for variance analysis.

Director
`PUT /budget-status { action: 'approve' }`





Phase B — Contracting & Procurement

Assign contractors, issue work orders, allocate BOQ, procure materials

▲

08

Contractor Selection & Work Order

Select contractors/sub-contractors from the Supplier module. Assign to BOQ items via `contractor` field. Define contractor unit prices separately from estimated prices. Work order = contract scope + rates.

PM + Costing Mgr
`BOQItem.contractor`
`batch-assign-contractor`



09

BOQ Allocation to Sub-Projects

Master BOQ items are allocated to block-level sub-projects with specific quantities. Tracks `allocatedQuantity` vs `remainingQuantity`. Prevents over-allocation beyond estimated quantity.

Costing Manager (MRM)
`POST /boq/allocate`



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Material Requisition (MR)

Site engineer raises MR for materials needed. This is MRM's "Material Requisition" input. Linked to specific BOQ items so material cost is tracked against budget.

Site Engineer
`Indents / MR module`



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Purchase Order (PO) → GRN → Material Issue

MR triggers PO creation from BOQ items. Vendor delivers → GRN recorded in Store. Materials issued to site. BOQ tracks `orderedQuantity` and `receivedQuantity`. Updates `totalCommitted` on project.

Procurement + Store
`POST /boq/create-po`
`PurchaseOrder model`





Phase C — Site Execution & Daily Monitoring

Track daily progress, record expenses, update task completion, site photos

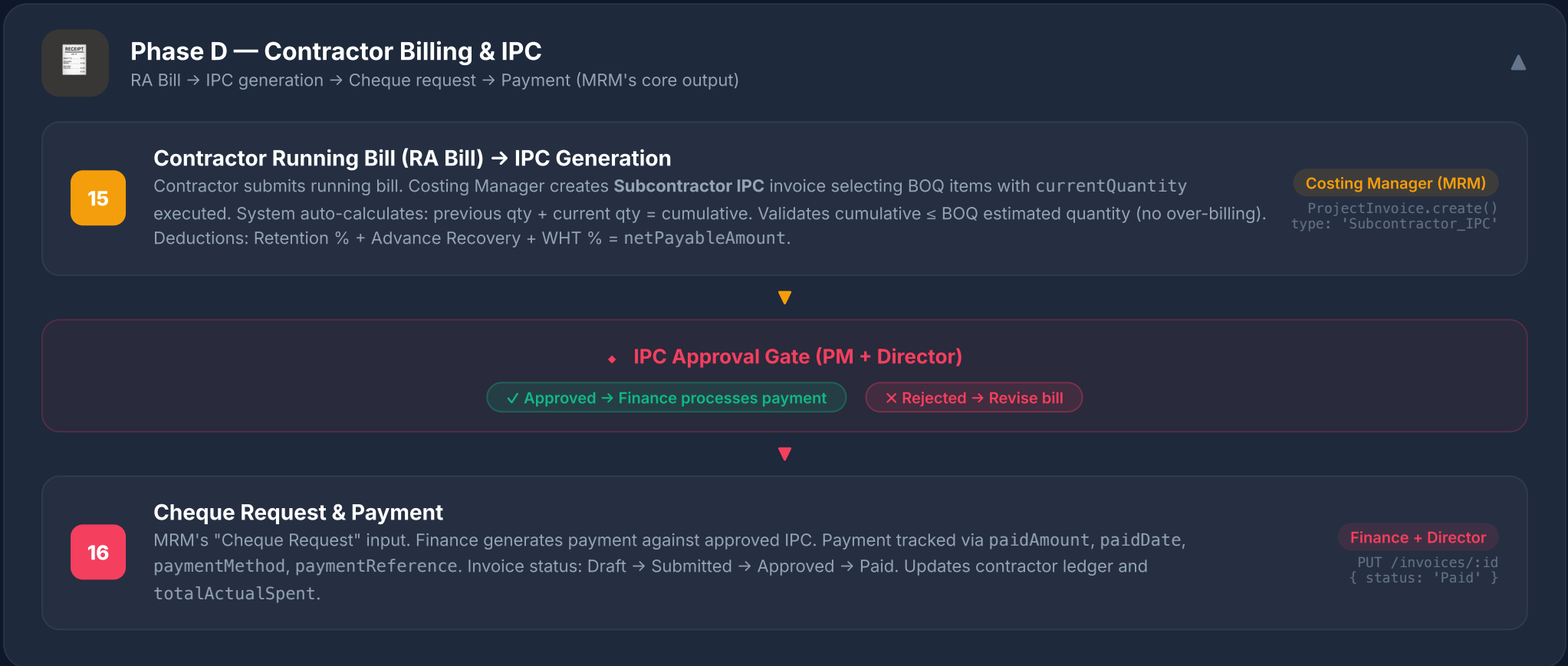
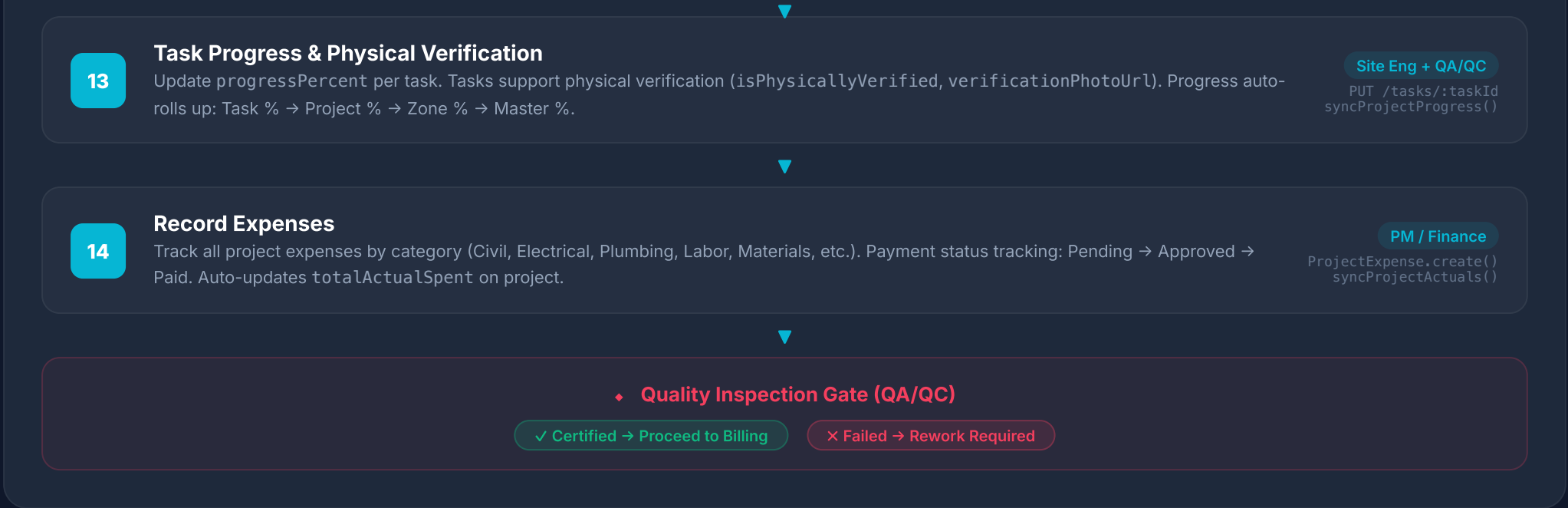
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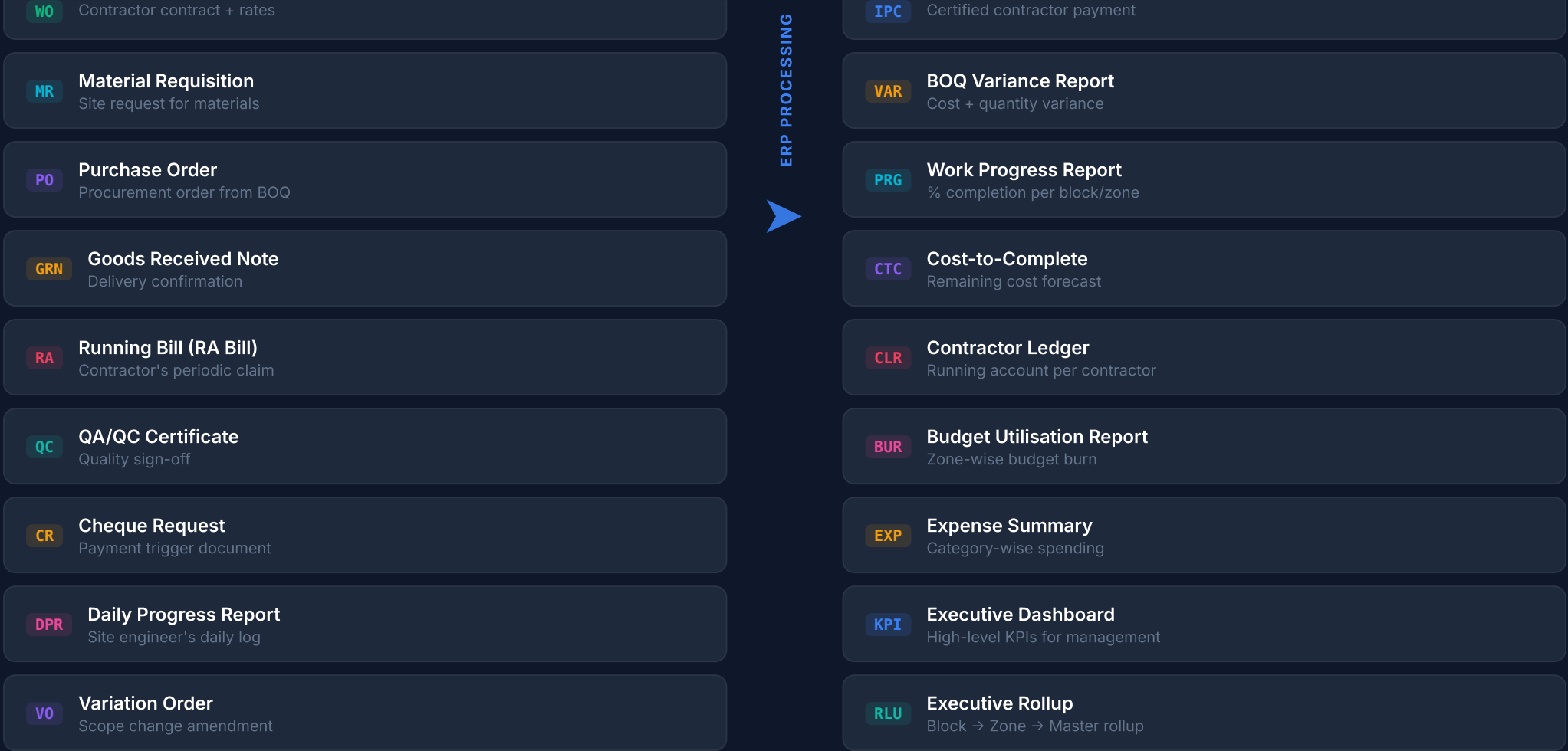
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Daily Progress Report (DPR)

Site engineer submits DPR with: weather, workforce count (civil, electrical, plumbing, supervisors), work done per task, materials used, issues/risks, and up to 20 site photos. MRM's DPR concept mapped directly.

Site Engineer
`DailyProgressReport.create()`
`POST /dpr (multipart)`

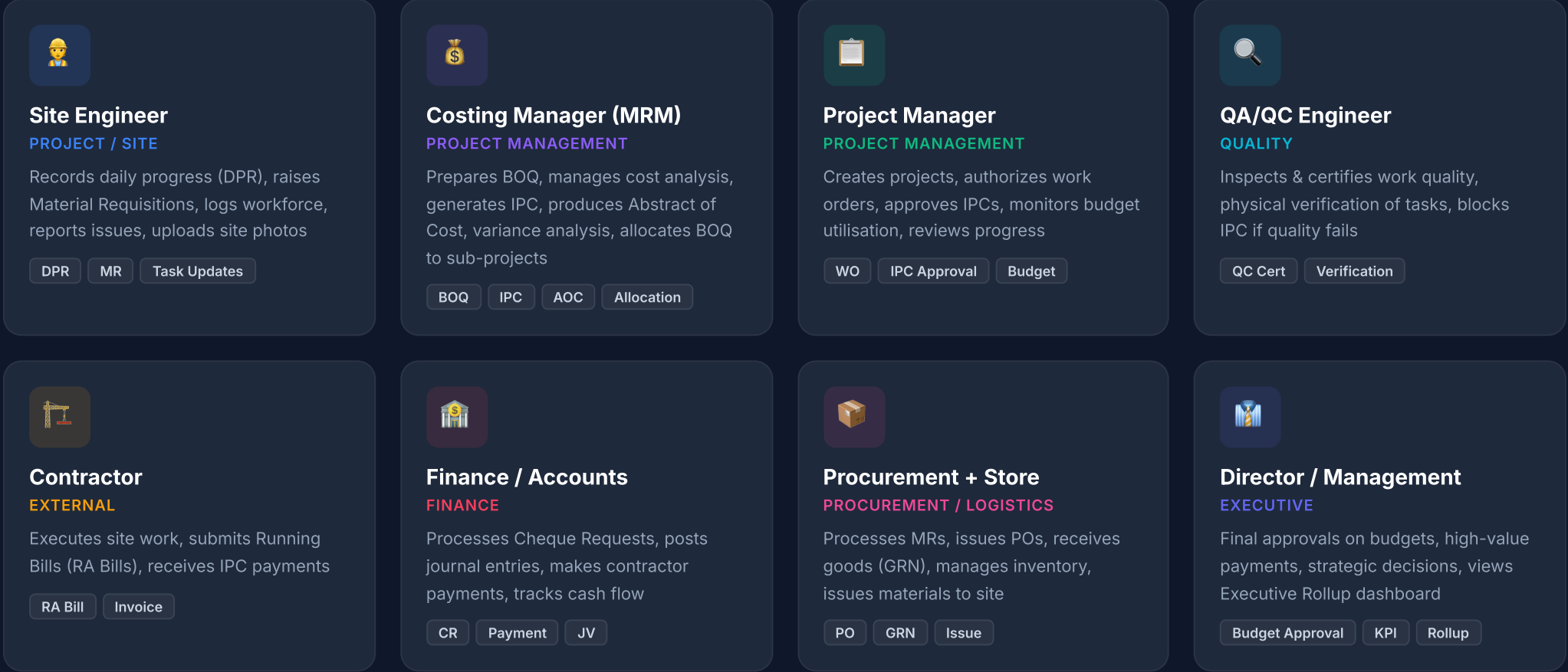




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Stakeholders & Responsibilities

Who does what at each step



MRM Concept	ERP Model	Key Fields	Status	Notes
IPC (Interim Payment Certificate)	ProjectInvoice	type: Subcontractor_IPC, retention, WHT	✓ Built	Auto-tracks cumulative billing with over-billing protection
Cheque Request	ProjectInvoice (status flow)	paidAmount, paidDate, paymentRef	🕒 Partial	Payment fields exist, dedicated CR module can be added
Purchase Order	PurchaseOrder	vendor, items, totalAmount	✓ Built	Created directly from BOQ items
Material Requisition	Indent (Stores module)	items, status, approvals	✓ Built	Via existing Stores/Indents module
Abstract of Cost	ConstructionProject	totalEstimated vs totalActualSpent	✓ Built	Auto-computed + Executive Rollup API
Budget Categories	ConstructionProject	budgetCategories[].estimatedAmount / approved	✓ Built	12 categories: Civil, Electrical, Plumbing...
Contractor Ledger	–	–	✗ Missing	Can be derived from ProjectInvoice grouped by contractor
Cash Flow Forecast	–	–	✗ Missing	Needs time-series projection from milestone dates

SGC ERP — Taj Residencia Project Management Module

MRM Costing Framework × ERP Integration Workflow | Generated September 2026